Beginner's Guide to Data Science Roles & Roadmap

1. Data Science Learning Roadmap

Level 1: Foundations
• Python Programming (Syntax, Libraries)
• Mathematics:
 - Linear Algebra
 - Probability
 - Statistics
• Tools:
 - NumPy
 - Pandas
 - Data Visualization (Matplotlib, Seaborn, Plotly)

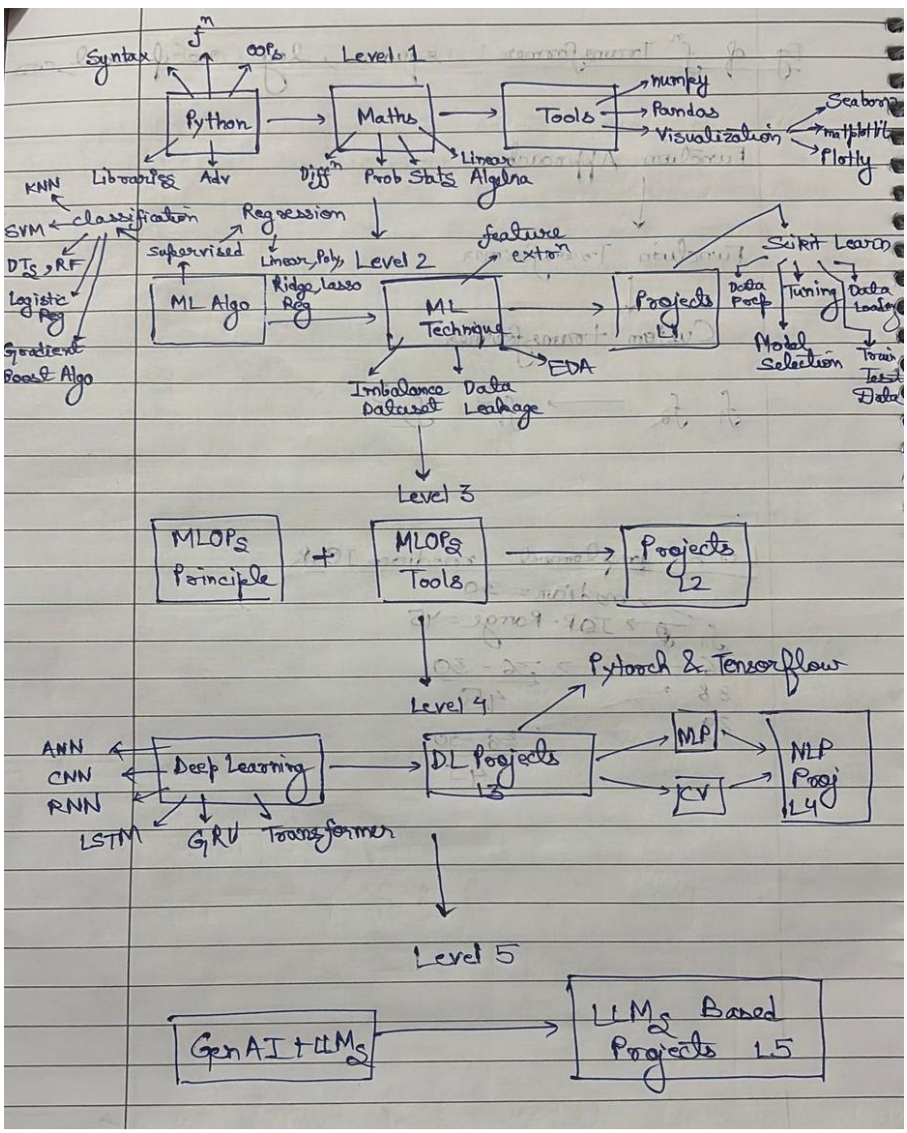
Level 2: Machine Learning
• Supervised Learning
• Regression
• Classification Algorithms:
 - KNN
 - SVM
 - Decision Trees
 - Random Forest
 - Logistic Regression
 - Gradient Boosting
• Important Concepts:
 - Feature Extraction
 - EDA (Exploratory Data Analysis)
 - Imbalanced Dataset Handling
 - Data Leakage

Level 3: MLOps
• MLOps Principles
• MLOps Tools
• Building scalable ML pipelines
• Model deployment and monitoring

Level 4: Deep Learning
• ANN (Artificial Neural Networks)
• CNN (Convolutional Neural Networks)

• RNN (Recurrent Neural Networks)
• LSTM / GRU
• Transformers
• Frameworks:
  - PyTorch
  - TensorFlow
• Projects:
  - NLP Projects
  - Computer Vision Projects

Level 5: Generative AI
• LLMs (Large Language Models)
• GenAI Applications
• LLM-based Projects



2. Major Job Roles in Data Science

1. Data Analyst
2. Data Engineer
3. Data Scientist
4. Machine Learning Engineer

3. Data Science Pipeline

Plan → Data Gathering (by data engineer) → Data Processing → EDA → Modeling → Evaluation → Deployment → Optimization

4. Data Engineer

Role:
A Data Engineer brings data from different sources so that other teams can use it efficiently.

Responsibilities:
• Scraping or extracting/pulling data from various sources
• Moving and storing data in optimal servers or data warehouses
• Building data pipelines and APIs for data access
• Managing databases and maintaining data infrastructure

Key Concepts:
• OLTP (Online Transaction Processing) – Basically website database
• OLAP (Online Analytical Processing) – Used for analytics and data warehouses

Data Engineer doesn't directly work on running databases as users interact with it. Data Engineers create OLAP from OLTP.

OLTP can be seen as database and OLAP can be seen as data warehouse

Skills Required:
• Strong understanding of Data Structures and Algorithms
• Programming: Python / Java / Scala
• Advanced DBMS
• Data Pipelines
• Distributed Systems
• Cloud Platforms (AWS, GCP, Azure)

• Big Data Tools (Apache Spark, Hadoop, Kafka, Hive)

5. Data Analyst

Role:
A Data Analyst analyzes and cleans data to generate insights and reports and summarizes the data.

Responsibilities:
• Cleaning and organizing raw data
• Analyzing data to derive insights
• Creating data visualizations
• Producing and maintaining reports
• Improving data collection procedures

Skills Required:
• Statistics
• Python / R
• SQL
• Excel
• Analytical thinking
• Data visualization
• Data storytelling
• Business understanding
• Communication skills

6. Data Scientist

Definition:
A Data Scientist is someone who is better at statistics than a software engineer and better at software engineering than a statistician.

Responsibilities:
• Building predictive models
• Working with large and unstructured datasets
• Applying advanced statistical techniques
• Using machine learning algorithms to forecast trends
• Solving complex business problems

Difference Between Data Analyst and Data Scientist:

• Data Analyst explains what is happening in the data.
• Data Scientist focuses on why it is happening and what will happen next.

Data Analyst focuses on understanding existing data to generate actionable insights, creates reports and supports business decisions using tools. In contrast, Data Scientist builds predictive models, works with large and often unstructured datasets and uses advanced statistical techniques and ML algos to forecast trends and solve complex problems.

7. Machine Learning Engineer

Role:
Once a model is created, the ML Engineer deploys it into production systems such as websites or applications. In simple words, ML Engineer integrates ready-made models into a website.

Responsibilities:
• Deploying machine learning models to production
• Scaling and optimizing models
• Monitoring model performance
• Maintaining deployed ML systems

Skills Required:
• Mathematics
• Python / Java
• Distributed Systems
• Machine Learning models
• Model evaluation
• Software engineering and system design

Comparisons

Role	Analytical Skills	Business Acumen	Data Storytelling	Soft Skills	Software Skills
Data Analyst	High	Medium to High	High	Medium to High	Medium
Data Engineer	Medium	Low	Low	Medium	High
Data Scientist	High	High	High	High	Medium
ML Engineer	Medium to High	Medium	Low	High	High